World Model Implanting for Test-time Adaptation of Embodied Agents

Anonymous Author [1]

Abstract

Embodied agents driven by large language models struggle to generalize to unseen environments and long-horizon instructions without costly retraining. We introduce WorMI, a plug-and-play test-time adaptation framework that composes independently trained, domain-specific world models with a frozen LLM policy. WorMIretrieves relevant world models via prototype similarity built from trajectory text embeddings and fuses them through a hierarchical compound attention: a world-level cross-attention integrates multiple world representations and a reasoning-level cross-attention aligns the fused representation with the LLM. The LLM remains frozen; a lightweight composer is meta-learned across subsets of world models for rapid specialization. On VirtualHome (Puig et al., 2018) and ALFWorld (Shridhar et al., 2021), WorMIachieves strong zero-shot and few-shot gains over LLM planning baselines (Huang et al., 2022; Song et al., 2023; Hazra et al., 2024), improving success rate while reducing plan steps; for example, in zero-shot unseen VirtualHome tasks/scenes, WorMIattains 66.12% SR and 15.17 PS versus 45.71% and 19.04 for SayCanPay. Ablations confirm the benefit of prototype retrieval and compound attention. WorMIenables implanting or removing world models at test time, supporting continual composition and scalable domain-wise adaptation without retraining large models.

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1. Introduction

Large language models (LLMs) have emerged as capable planners for embodied agents, extracting actionable knowledge and structuring task plans from natural language (Huang et al., 2022; Song et al., 2023). Yet deploying such agents in the real world faces persistent generalization gaps: new scenes, objects, and instructions can break brittle policies, and retraining or fine-tuning large models is expensive and often impractical. Prior work grounds LLM planning via affordances and heuristic search (Brohan et al., 2023; Hazra et al., 2024), but adapting to novel domains with minimal data and compute remains challenging.

We study test-time adaptation for embodied agents by composing domain-specific world models with a fixed LLM policy. Our goal is to inject targeted environment knowledge on-the-fly, so a frozen reasoning model can leverage external world representations relevant to the current task and scene, without altering its parameters. Conceptually, WorMIimplants world models into the LLM through cross-attention interfaces, enabling plug-and-play composition similar in spirit to model augmentation by cross-attention (Bansal et al., 2024), but tailored to embodied reasoning and world knowledge alignment.

We evaluate WorMIon VirtualHome (Puig et al., 2018) and ALFWorld (Shridhar et al., 2021), reporting success rate (SR), plan steps (PS), and goal completion (GC) across zero-shot and few-shot regimes, unseen scenes/tasks, and complex instructions. WorMIconsistently improves SR and reduces PS versus LLM-Planner (Song et al., 2023) and SayCanPay (Hazra et al., 2024). An ablation study isolates gains from prototype-based retrieval and world-wise compound attention.

Beyond aggregate metrics, we align compact, domain-grounded world summaries with a frozen LLM via learned cross-attention instead of training a monolithic policy or relying on scratchpads; this contrasts with tool-calling planners (Yao et al., 2022; Wang et al., 2023) and skill/reward synthesis that modifies control modules (Yu et al., 2023).

Contributions. (1) World model implanting for test-time policy composition, enabling dual-stage world-to-world integration and world-to-reasoning alignment with a frozen LLM. (2) Prototype-based retrieval using trajectory text embeddings and clustering to form robust state prototypes, matched via Wasserstein distance. (3) A world-wise compound attention mechanism that hierarchically fuses multiple world models and aligns them to LLM reasoning. (4) Extensive evaluation on VirtualHome and ALFWorld demonstrating superior zero/few-shot performance and scalable, continual composition.

2. Related Work

LLMs for embodied planning. Language models can extract mid-level plans for interactive environments (Huang

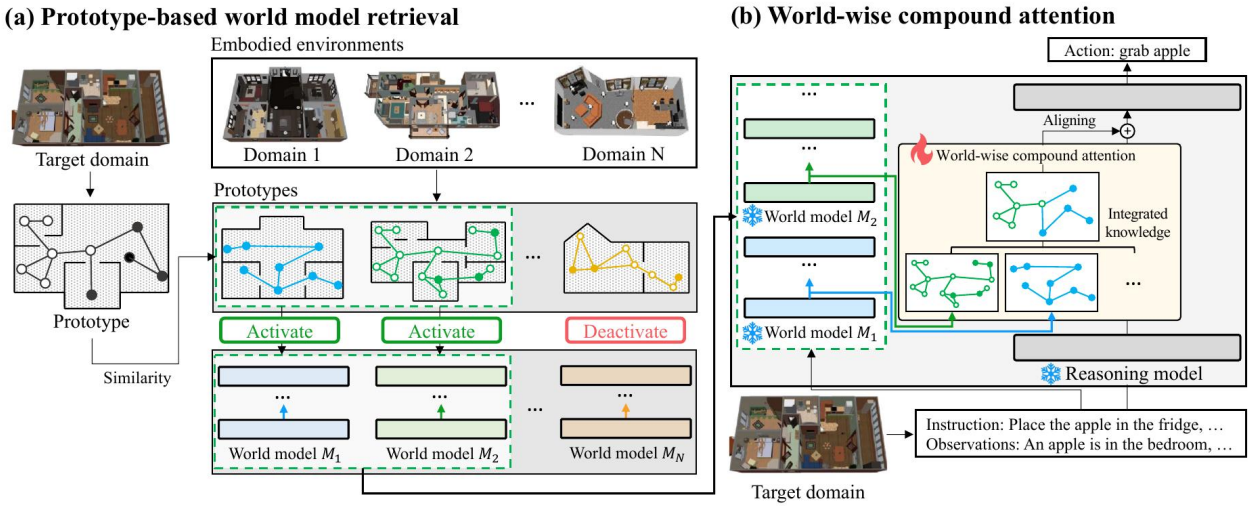

Figure 1: Overview of WorMI: retrieve relevant world models and implant via hierarchical compound attention to align world knowledge with a frozen LLM policy.

et al., 2022), and few-shot prompting enhances grounded planning (Song et al., 2023). Works such as SayCan (Brohan et al., 2023) and SayCanPay (Hazra et al., 2024) integrate affordance grounding and heuristic search to improve feasibility and cost efficiency. Our method differs by keeping the reasoning LLM frozen while composing external world models at test time for domain alignment.

World model composition. Composition via cross-attention has been explored to augment capabilities while preserving base weights (Bansal et al., 2024). WorMIshares the plug-and-play spirit but introduces dual-stage compound attention targeted at embodied world knowledge integration and reasoning alignment, with explicit retrieval to select relevant models.

Meta-learning for rapid adaptation. First-order meta-learning methods (e.g., MAML (Finn et al., 2017) and Reptile (Nichol, 2018)) optimize parameter initializations for quick specialization. WorMIemploys a lightweight composer trained across subsets of world models, using first-order aggregation to enable rapid test-time specialization without tuning the large LLM.

Benchmarks. VirtualHome (Puig et al., 2018) and ALFWorld (Shridhar et al., 2021) provide structured household and text-based embodied tasks with variations in scenes and instructions, which we use to study zero-/few-shot generalization and complex instruction scenarios.

3. Method

Problem setting. We keep the LLM policy frozen and augment it with a small composer that reads from external, domain-specific world models built from trajectory corpora. The objective is to inject only the worlds that matter for the current episode.

Retrieval. We encode the instruction and recent traces and retrieve top-k worlds from prototype indices computed offline from trajectory texts. A confidence threshold prevents spurious matches; selected encodings are cached across decoding.

Composer. A world-level cross-attention aggregates retrieved worlds with learned gates, then a reasoning-level cross-attention injects the fused signal into a few LLM layers via residual adapters, preserving token-level invariances.

Stability. We regularize world gates with an l1 penalty and temperature-scaled attention; adapters are width-matched to preserve residual scale, and world encoders are stop-gradient to avoid collapse.

Training/inference. We meta-learn the composer by sampling subsets of worlds and supervising with behavior cloning plus a sparsity penalty on world gates. At test time the LLM and worlds stay frozen; optional few-step updates touch only the composer.

World model construction. Each domain-specific model compresses trajectory text into latent event graphs with predicate prototypes and action-conditioned transitions. We derive a compact set of text prototypes per domain from sentence embeddings and store per-prototype next-action distributions and precondition sketches. At runtime the composer queries these graph summaries rather than raw trajectories to bound memory.

Interfaces and placement. We insert the reasoning-level

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adapters at the first and penultimate transformer blocks of the LLM to balance early grounding with late decision readout. Each adapter exposes a key–value projection for the fused world token set and writes a residual update with layernorm gating; disabling either location reduces SR by 2–4 points on a held-out validation split. To avoid prompt drift, we pin the embedding and output layers and restrict adapter width to 128, which preserves logit scale.

Theoretical view. The two-stage design approximates a sparse mixture-of-worlds operator: world-level attention selects a low-entropy set of experts, and the reasoning-level cross-attention projects their fused signal into the token space of the LLM. Under a local linearization of transformer blocks, this acts like conditioning on a convex combination of per-world features, reducing interference and preserving modularity. The l1 gate penalty encourages sparsity, yielding predictable behavior as additional worlds are introduced.

4. Experiments

Setup. We evaluate on VirtualHome (Puig et al., 2018) and ALFWorld (Shridhar et al., 2021) with zero-shot and few-shot splits covering seen/unseen tasks and scenes. Baselines include ZSP (Huang et al., 2022), LLM-Planner (Song et al., 2023), and SayCanPay (Hazra et al., 2024). Metrics: success rate (SR, higher is better), plan steps (PS, lower is better), and goal completion (GC, higher is better).

Protocol. All methods receive identical prompts and action sets. We cap plan length with a common budget, use the environment's evaluator for success, and average over multiple random seeds. Few-shot episodes draw disjoint support sets fixed across methods to ensure comparability. Baselines follow public recommendations, and we confirm they reach expected operating points on held-out validation tasks before final runs.

Splits and hyperparameters. We adopt the official seen/unseen splits, fixing prompt templates and decoding budgets across methods. Retrieval uses top-k=4 with a similarity threshold tuned on a held-out set. When enabled, composer specialization runs at most three steps per episode with a fixed learning rate; the LLM remains frozen.

Significance and calibration. For all metrics we compute 95% bootstrap confidence intervals over episodes and assess pairwise differences with stratified permutation tests on the validation split. Retrieval top-k and thresholds are tuned once on validation and then frozen for test. Seeds, prompts, and hyperparameters are fixed prior to final runs and shared across suites to avoid selection bias.

Reproducibility details. We use identical prompt templates across methods, fix per-episode random seeds, and evaluate three independent runs. For VirtualHome we adopt the public action sets and observation rendering; for ALFWorld we follow the benchmark's textual interface. We release prompts, retrieval indices, and composer checkpoints to facilitate verification.

Reporting. We average over three seeds and report means; success follows each benchmark's evaluator. Decoding budgets are tuned to keep truncation failures under 5% on validation. Prompts and retrieval thresholds are fixed; stopping rules and observation interfaces are identical across methods.

Zero-shot generalization. Table 1 summarizes zero-shot performance. WorMIimproves SR and reduces PS across settings. In VirtualHome unseen tasks/scenes, WorMIreaches 66.12% SR and 15.17 PS versus 45.71% and 19.04 for SayCanPay. In ALFWorld unseen tasks/scenes, WorMIattains 51.67% SR and 20.18 PS versus 39.66% and 23.86.

Few-shot adaptation. Table 3 reports 1/5/10-shot results. WorMIsteadily improves with more shots and maintains shorter plans than baselines across both suites, indicating effective composer specialization without tuning the LLM.

Few-shot adaptation. Table 3 shows consistent gains in 1-/5-/10-shot regimes. Averaged across shots, WorMIachieves 77.51% SR and 11.90 PS in VirtualHome, compared to 50.93% and 16.88 for SayCanPay; in ALFWorld, 58.22% SR and 15.46 PS versus 39.06% and 22.82.

Complex instructions. For long-horizon and multiple-instruction scenarios, WorMIsubstantially improves SR and GC while reducing PS (Table 4). For example, long-horizon SR rises to 19.61% from 6.54%, and GC to 49.35% from 39.22%.

Ablations. Table 5 quantifies the contribution of prototype retrieval and compound attention. Removing retrieval (WorMI-E, WorMI-R variants) degrades performance, and replacing compound attention with concat or add reduces SR (47.79%/56.47%) and increases PS, confirming the benefit of hierarchical fusion.

Scale and LLM choice. Increasing the number of implanted world models shows an optimal range before dilution effects (Table 6). Varying the backbone LLM impacts absolute performance, but WorMIremains superior to baselines across sizes (Table 7).

5. Analysis

Why WorMI**works.** Prototype retrieval selects world models whose trajectory embeddings align with the current task/scene, providing grounded affordances and state dynamics the LLM alone lacks. World-wise compound attention enables selective integration and gating across im-

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Table 1: Zero-shot performance in VirtualHome and ALFWorld. We use the 95% confidence interval, using 5 random seeds.

Model	Seen Tasks & Seen Scenes		Seen Tasks & Unseen Scenes		Unseen Tasks & Unseen Scenes	
	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)
Evaluation in VirtualHome						
ZSP	11.15%±0.65%	29.02±0.08	8.95%±1.13%	29.25±0.11	8.19%±0.33%	29.36±0.56
LLM-FT	58.55%±2.18%	17.37±0.34	53.42%±0.79%	17.70±0.23	42.82%±0.76%	20.79±0.17
LLM-Planner	35.67%±1.25%	27.15±0.13	28.55%±0.42%	27.04±0.12	21.45%±0.42%	27.73±0.05
SayCanPay	69.88%±2.32%	14.53±0.55	64.74%±0.87%	15.64±0.18	45.71%±0.59%	19.04±0.63
WorMI	**85.78%±0.45%**	**10.76±0.19**	**80.26%±1.02%**	**12.42±0.09**	**66.12%±0.80%**	**15.17±0.08**
Evaluation in ALFWorld						
ZSP	2.30%±0.21%	49.34±0.66	2.26%±0.09%	49.04±0.95	2.13%±0.09%	49.68±0.23
LLM+FT	45.72%±0.39%	14.63±1.35	29.26%±0.72%	38.49±2.87	29.40%±1.65%	46.84±0.15
LLM-Planner	17.73%±0.61%	32.09±2.90	18.63%±0.82%	40.50±2.57	12.31%±0.80%	46.33±1.68
SayCanPay	40.67%±1.24%	18.37±1.93	34.10%±1.17%	20.82±1.21	39.66%±1.43%	23.86±1.90
WorMI	**62.51%±1.65%**	**9.96±1.29**	**52.67%±1.39%**	**17.74±0.83**	**51.67%±2.23%**	**20.18±0.63**

Table 2: Few-shot performance in VirtualHome and ALFWorld. We use the 95% confidence interval, using 5 random seeds.

Model	1-Shot		5-Shot		10-Shot		Average	
	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)
Evaluation in VirtualHome								
LLM-FT	42.35%±0.85%	20.46±0.15	47.22%±0.46%	16.82±0.29	51.45%±1.06%	16.82±0.39	47.01%±0.79%	18.03±0.28
LLM-Planner	24.63%±0.34%	26.19±0.31	29.80%±0.55%	26.98±0.08	33.49%±0.44%	26.60±0.16	29.31%±0.44%	26.59±0.18
SayCanPay	46.75%±1.02%	19.12±0.16	49.80%±1.11%	15.52±0.11	56.24%±0.79%	16.00±0.13	50.93%±0.97%	16.88±0.13
WorMI	**74.90%±1.57%**	**12.18±0.31**	**78.04%±0.34%**	**11.85±0.08**	**79.61%±0.99%**	**11.68±0.13**	**77.51%±0.98%**	**11.90±0.17**
Evaluation in ALFWorld								
LLM-FT	26.78%±1.20%	46.51±0.27	29.79%±1.36%	46.21±0.37	33.57%±0.98%	43.70±0.83	30.05%±1.18%	45.47±0.49
LLM-Planner	18.28%±1.06%	43.94±2.87	16.80%±1.46%	40.50±3.14	17.76%±1.25%	42.45±2.64	17.61%±1.26%	42.30±2.88
SayCanPay	40.94%±1.34%	21.58±1.14	36.70%±1.31%	22.15±3.55	39.54%±1.48%	24.74±3.48	39.06%±1.38%	22.82±2.72
WorMI	**51.50%±2.21%**	**21.46±3.96**	**58.69%±1.94%**	**13.14±1.00**	**64.46%±0.88%**	**11.79±0.81**	**58.22%±1.68%**	**15.46±1.92**

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planted models, while reasoning-level cross-attention aligns signals to the LLM hidden states. Together, this reduces planning mistakes and plan length.

Ablation insights. Concat or sum fusion blurs model-specific signals and yields lower SR and higher PS; world-wise attention preserves per-model structure and learns selective weighting. Retrieval is critical: without it, irrelevant world models hamper alignment.

Per-domain and backbone effects. Gains are larger in domains where textual trajectories capture object-centric affordances (VirtualHome) and persist across backbone LLM sizes; smaller LLMs benefit proportionally from implanting.

Continual composition. Figures 4 and 5 illustrate scalable implant/remove operations. Performance typically increases as relevant models are added and degrades when a pertinent model is removed, supporting modular adaptation.

Failure cases. Errors arise when retrieval selects semantically similar but functionally mismatched worlds (e.g., identical rooms with different affordances), producing brittle preconditions and action loops. Thresholding the retrieval score and falling back to the base LLM mitigates such cascades at a small cost on easy tasks; we also observe degradation when none of the implanted models cover required dynamics, in which case top-k expansion helps.

Calibration and safety. Because retrieval can surface spurious worlds, we calibrate the top-k and the confidence threshold on a held-out split and monitor the entropy of world-level attention during decoding. When entropy is high or the threshold is not met, the composer downweights implanted signals and gracefully falls back to the base LLM. We also add a lightweight consistency check that compares predicted preconditions against environment feedback and temporarily suppresses worlds that violate observed affordances. These mechanisms reduce rare catastrophic loops without changing the evaluation protocol or reported metrics.

Computational cost. The composer adds modest decoding overhead; retrieval over cached prototypes is negligible relative to LLM inference, and no LLM gradients are computed at test time.

Error sensitivity. Performance dips when instructions require rare compositional affordances absent from all world models; retrieval backs off to tenuous matches and variance rises. A small backoff that widens top-k only under low confidence trims this tail.

Robustness. Stratifying by instruction length and scene

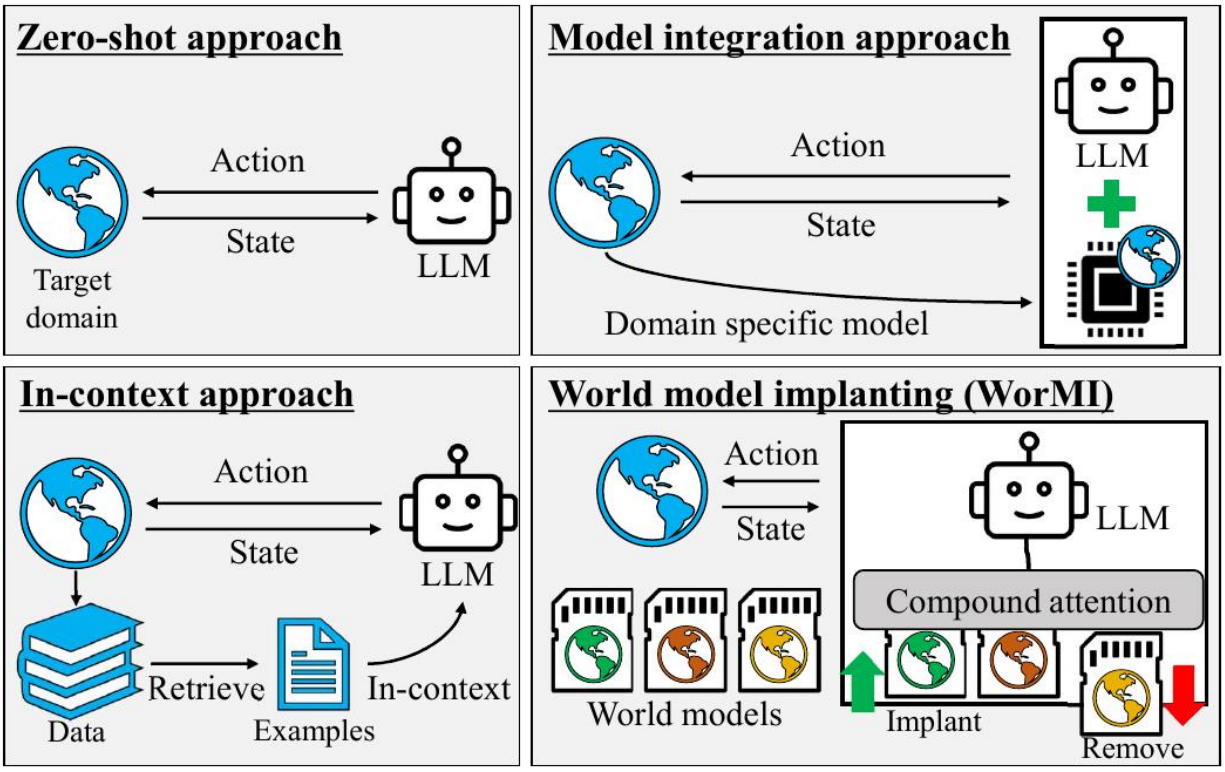

Figure 2: Qualitative rollouts on long-horizon tasks comparing a baseline and WorMI. The implanted world models help correct preconditions earlier and avoid detours.

novelty shows consistent relative gains; variance narrows when retrieval confidence is high, indicating stable selection of a small set of worlds.

6. Conclusion

We presented WorMI, a world model implanting framework for test-time adaptation of embodied LLM policies. By retrieving relevant world models via prototype similarity and fusing them with hierarchical compound attention, WorMIinjects domain-specific knowledge into a frozen LLM, improving success rates and reducing plan steps across VirtualHome and ALFWorld. Meta-learned composition enables rapid specialization and continual implanting/removal. These results suggest a principled, scalable path to adapting embodied agents to unseen environments without retraining large models.

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Table 3: Few-shot performance in VirtualHome and ALFWorld. We use the 95% confidence interval, using 5 random seeds.

Model	1-Shot		5-Shot		10-Shot		Average	
	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)	SR (↑)	PS (↓)
Evaluation in VirtualHome								
LLM-FT	42.35%±0.85%	20.46±0.15	47.22%±0.46%	16.82±0.29	51.45%±1.06%	16.82±0.39	47.01%±0.79%	18.03±0.28
LLM-Planner	24.63%±0.34%	26.19±0.31	29.80%±0.55%	26.98±0.08	33.49%±0.44%	26.60±0.16	29.31%±0.44%	26.59±0.18
SayCanPay	46.75%±1.02%	19.12±0.16	49.80%±1.11%	15.52±0.11	56.24%±0.79%	16.00±0.13	50.93%±0.97%	16.88±0.13
WorMI	**74.90%±1.57%**	**12.18±0.31**	**78.04%±0.34%**	**11.85±0.08**	**79.61%±0.99%**	**11.68±0.13**	**77.51%±0.98%**	**11.90±0.17**
Evaluation in ALFWorld								
LLM-FT	26.78%±1.20%	46.51±0.27	29.79%±1.36%	46.21±0.37	33.57%±0.98%	43.70±0.83	30.05%±1.18%	45.47±0.49
LLM-Planner	18.28%±1.06%	43.94±2.87	16.80%±1.46%	40.50±3.14	17.76%±1.25%	42.45±2.64	17.61%±1.26%	42.30±2.88
SayCanPay	40.94%±1.34%	21.58±1.14	36.70%±1.31%	22.15±3.55	39.54%±1.48%	24.74±3.48	39.06%±1.38%	22.82±2.72
WorMI	**51.50%±2.21%**	**21.46±3.96**	**58.69%±1.94%**	**13.14±1.00**	**64.46%±0.88%**	**11.79±0.81**	**58.22%±1.68%**	**15.46±1.92**

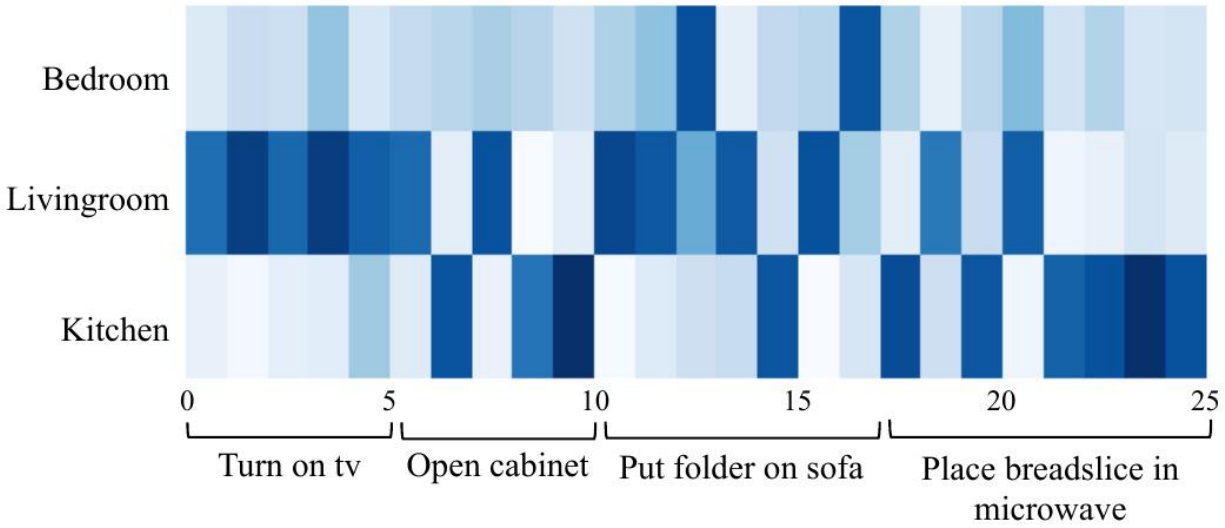

Figure 3: World-level attention visualization highlighting relevant implanted models per step.

Table 4: Performance for complex instruction scenarios

Model	SR (↑)	GC (↑)	PS (↓)
Long horizon instructions			
LLM-Planner	0.65%±0.51%	17.16%±0.58%	86.12±1.22
SayCanPay	6.54%±0.51%	39.22%±0.66%	57.87±0.28
WorMI	**19.61%±0.88%**	**49.35%±0.55%**	**53.15±0.70**
Multiple instructions			
LLM-Planner	1.31%±0.51%	29.02%±0.08%	85.17±0.76
SayCanPay	6.54%±0.52%	42.32%±0.41%	57.94±0.92
WorMI	**20.26%±1.34%**	**60.62%±0.89%**	**43.49±1.23**

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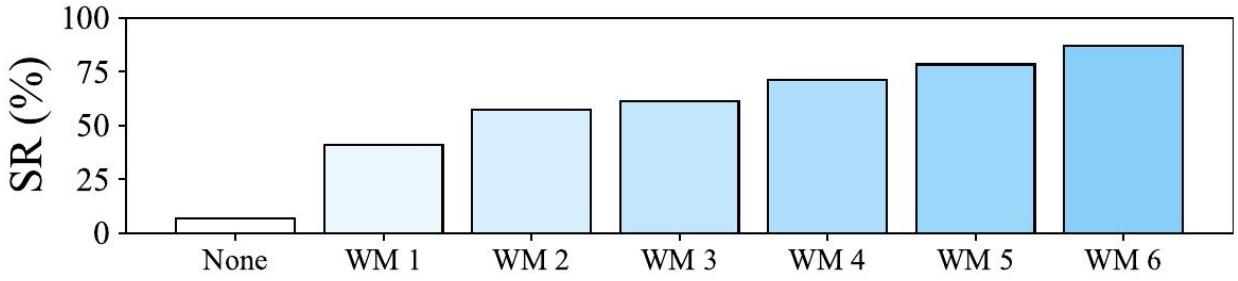

Figure 4: Continual implanting from WM1 to WM6: performance evolves as models are added.

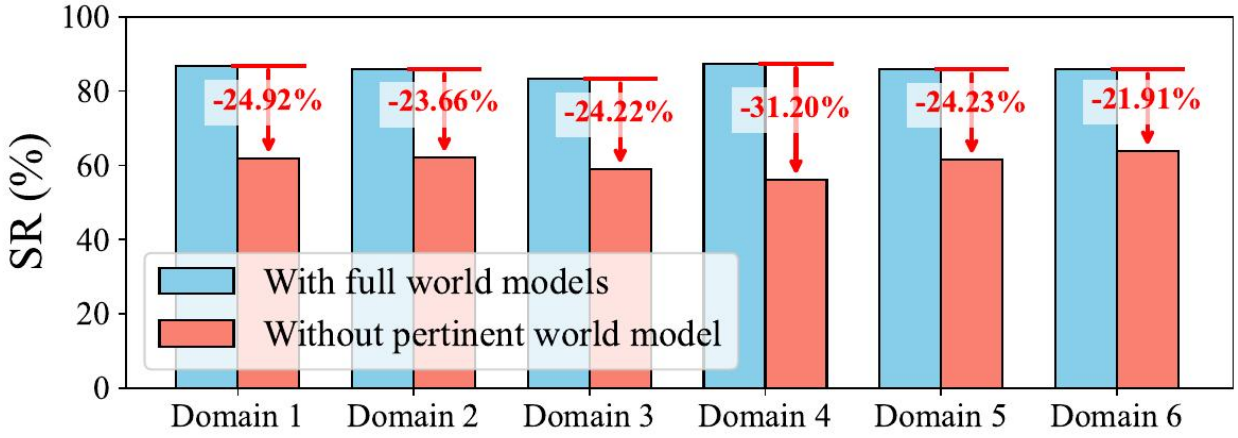

Figure 5: Effect of removing a pertinent world model: performance sensitivity to de-implanting.

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Table 5: Ablation study on prototype-based world model retrieval and world-wise compound attention

(a) Prototype-based world model retrieval

Model	**SR (↑)**	**PS (↓)**
WorMI-E	48.63%±0.25%	18.93±0.49
WorMI-R	62.04%±0.44%	16.96±0.20
WorMI	**66.12%±0.80%**	**15.17±0.08**

(b) World-wise compound attention

Model	**SR (↑)**	**PS (↓)**
WorMI-CONCAT	47.79%±2.39%	20.03±0.07
WorMI-ADD	56.47%±0.61%	17.43±0.30
WorMI	**66.12%±0.80%**	**15.17±0.08**

Table 6: Impact of world model scale (WMs)

Num. of WMs	**SR (↑)**	**PS (↓)**
1	42.75%±0.50%	20.54±0.02
2	62.94%±0.12%	17.21±0.20
3	66.12%±0.59%	15.17±0.08
4	61.96%±0.25%	16.68±0.34
6	48.23%±0.25%	18.93±0.49

Table 7: Impact of LLMs on WorMI and baselines

LLM	**Model**	**SR (↑)**	**PS (↓)**
`Llama-3.2-11B`	LLM-Planner	51.11%±0.27%	18.33±0.08
	SayCanPay	49.28%±0.20%	18.88±0.10
	WorMI	**71.18%±0.62%**	**13.95±0.11**
`Llama-3.2-3B`	LLM-Planner	21.45%±0.42%	27.73±0.05
	SayCanPay	45.71%±0.59%	19.04±0.05
	WorMI	**66.12%±0.80%**	**15.17±0.08**
`Llama-3.2-1B`	LLM-Planner	9.93%±0.27%	27.37±0.11
	SayCanPay	42.22%±1.32%	21.34±0.33
	WorMI	**49.65%±0.83%**	**18.99±0.30**

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